

Improving Clinic Efficiency Through Data Analysis at Hometown Medical PLLC

Fayetteville State University

BIDA 670: Advanced Analytics Project

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Executive Summary

Hometown Medical Professional Limited Liability Company (PLLC), an independent primary care clinic in Lillington, North Carolina, faced increasing financial pressure driven by declining reimbursement rates and rising operating costs. Although revenue grew from approximately \$121,000 in 2021 to over \$422,000 in 2023, net income remained near zero, indicating that the clinic's volume-driven model is no longer sustainable (Hometown Medical PLLC, 2026c). The analysis identified three primary drivers of this financial strain, namely low revenue per encounter, significant referral leakage, and administrative inefficiencies that limited clinical capacity.

The project proposed a strategic shift toward a value-driven model intended to increase revenue per encounter rather than patient volume. Key recommendations included expanding in-house diagnostic capabilities through Point-of-Care Ultrasound (POCUS) and implementing value-based services such as Chronic Care Management (CCM) and Annual Wellness Visits (AWV). The plan also outlined workflow optimizations and revenue cycle improvements designed to raise the clinic's net collection rate from its current 87 percent toward industry benchmarks.

Ultimately, these combined strategies propose a practical framework for improving financial performance while reducing reliance on external referrals. By capturing previously lost revenue and improving operational efficiency, the clinic can support long-term sustainability without requiring continued growth in patient volume. This approach supports the clinic's ability to maintain clinical independence while continuing to deliver high-quality patient care within a constrained reimbursement environment.

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I. INTRODUCTION

a. Context & Scope

This project evaluates the operational and financial challenges facing Hometown Medical Professional Limited Liability Company (PLLC) with a focus on long-term sustainability. The practice is physically located in Lillington, North Carolina, a low-income area in Harnett County, where the poverty rate stands at around 14% (World Population Review, 2026). This local economic context is relevant because the clinic often serves patients who may be uninsured, underinsured or covered through public programs such as Medicare and Medicaid. These factors directly influence both patient demand and reimbursement patterns within the clinic's operating environment.

In response to continued growth, Hometown Medical PLLC relocated to a larger, more modern facility to increase clinical capacity and support a higher patient volume. This expansion aligns with the clinic's goal of growing its patient panel while maintaining access to care within the community. Patient demand is also influenced by broader coverage changes, including North Carolina's Medicaid expansion, which brought more than 650,000 newly eligible residents into the healthcare system statewide (North Carolina Department of Health and Human Services, 2025). The new facility also supports the delivery of additional services, including primary care and advanced laser treatments for hair removal or skin rejuvenation.

The primary challenge facing the clinic is margin compression driven by constrained reimbursement rates and rising operating costs (Centers for Medicare and Medicaid Services, 2025). At the same time, limited reimbursement per encounter creates a revenue squeeze, requiring higher patient volume to sustain overall revenue (Kaiser Family

Foundation, 2026). As a result, the volume-driven model becomes less effective, as additional visits do not proportionally improve financial performance. In response to these constraints, this analysis examines how Hometown Medical can move beyond high-volume primary care by incorporating higher-value services, including in-house diagnostic imaging such as Point-of-Care Ultrasound (POCUS). Using data analysis techniques, this project provides a framework for improving profit margins by increasing revenue per patient encounter rather than relying on continued growth in patient volume.

b. Background & Policy Environment

Hometown Medical PLLC operates within a complex healthcare landscape where reimbursement structures often favor large health systems over independent providers. A primary example of this imbalance is the use of facility fees, which help hospitals and hospital-affiliated outpatient clinics cover overhead and care team support costs. These fees can result in higher total reimbursement for services delivered in hospital outpatient settings compared with similar services provided by independent physician offices. Independent practices generally cannot bill separate facility fees, which places them at a competitive disadvantage in covering the full cost of care delivery (Dobson, 2026).

These structural disparities are further compounded by constrained and evolving reimbursement rates, which place additional pressure on the clinic's single-provider model. For instance, UnitedHealthcare implemented a 15% payment reduction for certain services provided by advanced practice providers, including physician assistants, unless those services are performed in collaboration with a physician and billed according to applicable requirements (UnitedHealthcare, 2025). Additionally, the Centers for Medicare and Medicaid Services (CMS) finalized a 2.83% decrease in the 2025 Physician Fee Schedule

conversion factor, which determines how relative value units (RVUs), a measure used to assign value to medical services and calculate payment amounts, are applied (Centers for Medicare and Medicaid Services, 2025).

Given that the practice relies on a single physician assistant for most clinical revenue, these changes have a direct and disproportionate impact on the organization's financial sustainability. As healthcare consolidation continues across North Carolina, the practice must manage rising labor costs and administrative demands while operating within constrained reimbursement structures to remain independent (Leslie, 2022). This independence ensures patients continue to receive personalized, community-based care tailored to local needs.

c. Motivation & Goals

The motivation for this project is to assist Hometown Medical PLLC in identifying operational inefficiencies and evaluating solutions that support long-term financial sustainability. The primary objective is to analyze clinic data to determine how reimbursement constraints and administrative burdens affect the revenue cycle. This includes identifying areas of revenue leakage and assessing whether service diversification can offset declining primary care margins. Specifically, the analysis evaluates whether incorporating diagnostic tools such as POCUS can provide a sustainable revenue stream within a low-income patient population. Rather than relying on discretionary services, the project prioritizes care that is deemed medically necessary and aligns with the clinic's payer mix and community needs. Ultimately, the goal is to provide actionable insights that improve resource utilization and support the clinic's ability to remain a financially viable independent practice.

II. PROBLEM DEFINITION

Hometown Medical PLLC faces a significant threat to its long-term viability due to rising operational costs and declining reimbursement rates. The current business model relies heavily on high-volume primary care visits, which are increasingly insufficient to sustain profitability. Preliminary financial data indicates the clinic collects approximately 50 to 55 percent of its billed charges due to insurance and government reimbursement structures (Hometown Medical PLLC, 2026f). This gap between billed charges and collected revenue limits the clinic's ability to convert patient volume into sustainable financial performance and contributes to diminishing returns as patient volume increases. This overreliance on volume places significant structural pressure on the practice, which depends on a single physician assistant for most of its clinical revenue.

These financial constraints are further intensified by the clinic's patient population. The clinic serves a low-income community, which limits growth from elective or high out-of-pocket services. This dynamic reinforces dependence on primary care visits with constrained reimbursement rates and increases vulnerability to future payment reductions. Consequently, the clinic has limited flexibility to offset declining reimbursement through alternative revenue streams.

Internal data reveals significant revenue leakage driven by a wide gap between billed charges and collections, as well as high out-referral rates for services that could be managed in-house. These patterns highlight a fundamental misalignment between the clinic's current service model and the realities of the reimbursement landscape. Consequently, the clinic is missing critical opportunities to optimize its financial performance and maximize the revenue generated per patient encounter.

Combined, these pressures indicate that the clinic’s core challenge is not patient demand, but the ability to generate sufficient revenue per encounter. This distinction shifts the focus from increasing patient volume to improving the value derived from each patient interaction. Accordingly, this project evaluates whether restructuring the clinic’s service mix can increase revenue per encounter and improve financial sustainability without relying on continued growth in patient volume.

III. LITERATURE REVIEW

The financial sustainability of independent primary care practices has become more challenging under evolving reimbursement models, rising costs, and increasing administrative demands. Although the Affordable Care Act and Medicaid expansion improved insurance coverage and access to care, these changes have also introduced financial tradeoffs for providers (Creedon et al., 2022). Expanded coverage has increased patient demand, yet reimbursement rates, particularly from Medicaid and Medicare, often remain insufficient to cover the full cost of care delivery, contributing to persistent margin pressure for independent practices (Donohue et al., 2022; Kaiser Family Foundation, 2026). These pressures reflect a broader shift toward value-based care, where providers are increasingly rewarded for quality outcomes and efficiency rather than service volume (Johns Hopkins ACG System, 2023). These dynamics are central to this project, which examines how these structural pressures affect the financial and operational performance of an independent primary care clinic.

Recent policy and payer-level changes have further intensified these pressures. Adjustments to the Medicare Physician Fee Schedule and reductions in reimbursement for services provided by advanced practice providers have directly constrained revenue growth

(Centers for Medicare and Medicaid Services, 2025; UnitedHealthcare, 2025). At the same time, structural disparities in reimbursement, such as the availability of facility fees for hospital affiliated systems, continue to disadvantage independent practices, which lack access to these supplemental revenue mechanisms (Dobson, 2026). These dynamics disproportionately affect smaller clinics operating without the scale or negotiating leverage present in larger health systems.

Beyond reimbursement constraints, the administrative complexity of healthcare delivery has emerged as a significant operational burden. Increasing requirements for prior authorizations, documentation, and billing compliance have expanded nonclinical workloads, driving up overhead costs and limiting provider productivity. Research in healthcare operations and system efficiency suggests that these administrative demands weaken the relationship between patient volume and financial performance by limiting the efficiency with which clinical activity is converted into revenue (American Hospital Association, 2022). These challenges are particularly pronounced in small independent clinics, where limited staffing and financial resources amplify the impact of reimbursement and administrative pressures.

At the same time, the limitations of volume driven care models have become more apparent. While increased patient throughput can drive revenue growth, it often results in diminishing marginal returns due to capacity constraints, staffing limitations, and rising variable costs. This is particularly evident in rural and semi-rural settings, where provider shortages and resource constraints amplify operational inefficiencies (Lindrooth et al., 2018). In response to these constraints, recent literature has shifted focus toward improving revenue per patient encounter rather than relying solely on volume expansion (American Academy of Family Physicians, 2023).

These findings collectively suggest that improving revenue per patient encounter, rather than increasing patient volume, may be critical for long-term sustainability. Despite these insights, limited research examines how these structural and policy-driven pressures translate into day-to-day financial performance and operational decision making at the individual clinic level. Much of the existing research relies on aggregated national or state-level data, which can obscure organization-specific inefficiencies and opportunities for improvement (Buchmueller et al., 2024). This gap is particularly relevant for independent clinics, where operational decisions directly impact financial sustainability.

This project addresses that gap by examining a single independent primary care clinic and evaluating how reimbursement pressures, workflow inefficiencies, and service limitations interact to shape financial outcomes. The analysis integrates internal financial and operational data with external benchmarks to provide a more detailed view of performance drivers. By linking broader healthcare trends to organization-level data, the project offers practical insight into how independent practices can improve sustainability through targeted operational improvements and strategic service expansion.

IV. APPROACH / METHODOLOGY

This project adopted a single-organization case study design to evaluate the operational and financial performance of Hometown Medical PLLC within the broader context of healthcare reimbursement pressures. A case study approach was appropriate given the need to examine real-world organizational behavior, financial constraints, and workflow inefficiencies within a specific and bounded system (Yin, 2018). This design allowed for a focused analysis of

how declining reimbursement rates for mid-level providers interacted with internal workflows to affect the clinic's long-term financial sustainability.

To ensure a comprehensive evaluation, the project employed a mixed-methods approach that integrated both quantitative and qualitative data (Creswell & Creswell, 2018). Quantitative analysis focused on financial and operational metrics, including reimbursement trends, provider productivity, cost structures, and referral patterns for ancillary services. These metrics corresponded directly to internal datasets, including financial records, appointment data, and referral data collected from the clinic. In parallel, qualitative data was gathered through structured questionnaires completed by clinical staff and billing administrators, alongside workflow observations. These methods were used to identify patterns in revenue performance, operational inefficiencies, and areas of potential revenue loss.

The analytical framework was guided by healthcare operations management principles and project performance evaluation techniques, particularly those related to efficiency, resource utilization, and revenue cycle optimization. Key concepts included workflow efficiency, capacity utilization, and cost-to-serve analysis, which are used to evaluate how effectively the clinic converted clinical activity into revenue. These concepts were applied to assess the relationship between patient volume, operating costs, and revenue per encounter.

To further structure the analysis, a gap analysis approach was used to compare the clinic's current practices against established best practices in ambulatory care management and revenue cycle optimization. This comparison evaluated the clinic's performance relative to recognized benchmarks, such as those from the Medical Group Management Association (MGMA) and the American Medical Association (AMA), with a focus on provider

productivity, revenue capture, and the integration of ancillary services to identify specific gaps in the clinic's current operational model.

This structured methodology ensured that all recommendations were aligned with standard industry best practices and directly supported the clinic's long-term financial viability. By linking internal operational data with external benchmarks, the project provided a practical framework for improving revenue per patient encounter. It also supported the evaluation of in-house diagnostic expansion, including POCUS, through financial feasibility and referral pattern analysis.

V. DATA SOURCES & COLLECTION

Data collection for this project relied on a combination of primary and secondary sources to ensure both analytical depth and methodological rigor. Integrating multiple data types supported a comprehensive evaluation of the clinic's operational and financial performance by strengthening the consistency and reliability of the findings. External benchmarks from industry sources, including CMS, MGMA, and AMA, were used to contextualize the clinic's performance and support comparative analysis.

(1) Primary Data Collection: Primary data were collected during an on-site visit to Hometown Medical PLLC through direct observation, internal document review, and structured data collection using a standardized worksheet completed with input from the clinic owner. This process captured key information related to clinic operations, financial performance, referral patterns, and administrative workload.

- **Structured Data Collection:** A guided worksheet was used to collect both quantitative inputs and qualitative responses from the clinic owner. This included information on patient

volume, referral patterns, administrative workload, and operational challenges. These responses provided context for interpreting internal datasets and supported the overall analysis.

- **Direct Observation:** This involved an on-site assessment of clinic workflows and patient flow. These observations provided real-world context for the administrative bottlenecks and operational challenges later identified in the hard financial and appointment data provided by the clinic.

(2) Internal Document and Data Review: A critical component of the analysis involved reviewing internal documents and detailed spreadsheets provided by the clinic, which served as the foundation for quantitative evaluation. These datasets were exported directly from the clinic's management systems and included financial reports, appointment records, referral logs, and reimbursement data.

- **Financial & Revenue Data:** Data included reimbursement rates by payer type, revenue trends over time, and transaction-level financial reports. These data supported evaluation of revenue variability and the financial impact of payer mix.
- **Productivity & Utilization:** Operational and productivity data included patient volume, visits per day, and detailed records of cancelled and missed appointments extracted from internal scheduling spreadsheets, which were used to assess clinic capacity and utilization.
- **Referral & Leakage Data:** Referral data included the volume and types of services sent to external providers. These data were used to identify patterns of revenue leakage and evaluate opportunities for service retention.
- **Operational Efficiency & Cost Structure:** Additional input included administrative workload, staffing structure, and time spent on non-clinical tasks such as prior

authorizations and documentation. These data supported evaluation of operational efficiency and cost-to-serve.

(3) Secondary Data Sources: Secondary data provided context and established benchmarks for comparison. These sources included publicly available reimbursement data from CMS, as well as industry benchmarks from MGMA and AMA related to primary care productivity and revenue cycle performance. Peer-reviewed literature and reports from organizations such as the Kaiser Family Foundation (KFF) were also used to provide insight into broader healthcare trends and policy impacts. These sources enabled comparison between the clinic's performance and established industry standards to identify areas for improvement. This ensured that recommendations were realistic and aligned with industry performance standards.

VI. RESULTS & ANALYSIS

The analysis integrates internal financial and operational data from Hometown Medical PLLC with supporting benchmarks to evaluate whether the clinic's current high-volume primary care model is financially sustainable. The findings are organized across key financial and operational dimensions. The results indicated that while patient volume and total revenue increased significantly, profitability did not improve proportionally with revenue growth. This suggests structural inefficiencies and limitations within the current service model that hinder the clinic's ability to capitalize on its existing patient base.

(1) Financial Performance Trends

A review of financial data from 2021 through 2023 reveals that while revenue is growing, profitability is not improving at the same pace. This divergence indicates that increased patient volume alone is not translating into sustainable financial performance over

time. Instead, the data suggests that additional volume is accompanied by proportional or greater increases in operating costs. For instance, revenue increased from approximately \$121,886 in 2021 to \$422,723 in 2023, representing an increase of approximately 247% over a two-year period. However, despite this growth, net income in 2023 remained effectively near zero, indicating that increased patient volume has not translated into improved profitability (Hometown Medical PLLC, 2026c). This divergence further confirmed that the clinic's current volume-driven model was structurally inefficient and unable to convert growth into sustainable financial gain.

Internal financial reporting for the first quarter of 2026 further highlights the significant gap between billed charges and realized revenue. For the reporting period ending March 31, 2026, total charges exceeded \$178,000 with contractual adjustments accounting for more than \$92,000 in reduced revenue (Hometown Medical PLLC, 2026f). This disparity reflects a significant gap between billed charges and realized revenue within the reporting period and highlights the impact of contractual adjustments and claims processing on revenue conversion. These figures reinforce how current reimbursement models limit the clinic's ability to convert high patient volume into sustainable financial performance.

(2) Cost Structure and Margin Compression

An analysis of operating expenses highlights the primary drivers of margin compression. Total overhead expenses increased significantly, reaching approximately \$235,871 in 2023, with labor-related costs, including salaries and compensation, comprising a substantial share of overall expenses. Billing-related expenses alone reached approximately \$54,975, representing a significant share of total overhead and highlighting the administrative burden associated with revenue collection (Hometown Medical PLLC, 2026c). This level of billing expense indicates

that a considerable share of revenue is consumed by the processes required to code, submit, and collect claims, reducing efficiency and delaying cash flow realization. Additional cost pressures included medical supplies, equipment-related expenses, and facility overhead. These data points indicated that the clinic's cost structure increased with activity levels, which limited its ability to improve margins. The clinic did not appear to realize the benefits associated with economies of scale, as increased activity led to proportional cost increases rather than improved efficiency.

(3) Volume-Based Model Limitations

Operational data indicated that the clinic was already functioning within a high-volume care model with approximately 16 to 20 patients seen per day. While this level of throughput supported revenue growth, it also introduced capacity constraints such as limited provider availability and an increased administrative burden. Consequently, internal data indicated that each additional visit contributed relatively low incremental revenue (Hometown Medical PLLC, 2026a, 2026c). These findings confirmed that the clinic was operating at or near full capacity, making further volume-based growth an ineffective strategy for improving financial performance. This reinforces the need for a shift in the underlying service model.

(4) Revenue Efficiency and Value per Encounter

A central finding of this analysis was the disconnect between patient volume and revenue efficiency. This constraint was driven by reimbursement rates tied strictly to primary care services and a payer mix heavily influenced by Medicare and Medicaid. Consequently, the lack of higher-value ancillary services prevented the clinic from optimizing the financial return on its existing patient base, as reflected in internal financial data (Hometown Medical PLLC, 2026c).

Publicly available Medicare service-level data from CMS further illustrates this constraint by showing differences in reimbursement across service categories. Routine office visits such as CPT 99213 and 99214 generate moderate reimbursement, while diagnostic services such as CPT 76882 can provide an additional billable service when performed in a non-facility setting (Centers for Medicare and Medicaid Services, 2026). Integrating diagnostic services into primary care visits allows multiple billable components within a single encounter, increasing total revenue without requiring additional patient volume. Improving revenue per visit therefore depended on expanding in-house capabilities that aligned with existing patient demand.

This highlights the importance of service mix as a primary driver of financial performance rather than patient volume alone. Without higher-value services integrated into the care model, the clinic remains constrained by the reimbursement ceiling of primary care visits. Improving revenue per encounter therefore depends on expanding in-house capabilities that align with existing patient demand.

(5) Referral Patterns and Revenue Leakage

A detailed review of referral data provided by Hometown Medical PLLC further supports the presence of revenue leakage within the clinic's current operating model. The dataset includes more than 1,100 outbound referrals recorded between November 11, 2025, and April 10, 2026, averaging approximately 150 referrals per month across a range of service categories, including diagnostic imaging and specialty care, based on available internal reporting and manual aggregation (Hometown Medical PLLC, 2026d). This volume indicates that a substantial portion of patient care extends beyond the clinic's internal capacity and capability.

Analysis of these referrals showed that many services were directed to external providers rather than being retained within the clinic. Many of these services involved diagnostic or procedure-based care that, while not currently offered in-house, may represent potential opportunities for internal evaluation based on referral volume and reimbursement structure. To illustrate this leakage, musculoskeletal (MSK) ultrasound referrals were analyzed as a representative example. The data showed approximately 80 referrals for MSK ultrasounds (CPT 76881 and 76882) over a seven-month period. According to the CMS Physician Fee Schedule, the national average non-facility reimbursement for a limited MSK ultrasound (CPT 76882) was approximately \$95 (Centers for Medicare and Medicaid Services, 2026). Capturing an estimated 50% of these referrals in-house could generate approximately \$3,800 in additional revenue from this service line alone (Hometown Medical PLLC, 2026d). This pattern indicates that revenue leakage is not isolated to a single diagnostic category but reflects a broader structural gap in service capacity.

While some referrals remained clinically necessary, the volume and distribution of outbound services indicated that select diagnostic procedures were strong candidates for in-house evaluation. Retaining even a portion of these services would increase revenue per patient encounter without requiring a further increase in patient volume. This strategic shift had the potential to significantly improve financial performance by capturing revenue that was currently lost to external providers.

(6) Operational Efficiency Constraints

Workflow and staffing structure further contribute to the clinic's financial limitations. The practice operates with a lean staffing model, including a single primary provider supported by medical assistants and administrative staff. Administrative responsibilities such as prior

authorizations and documentation consume approximately four hours per day, significantly reducing provider capacity for revenue-generating clinical activities (Hometown Medical PLLC, 2026b). Workflow delays are most pronounced during check-in and intake processes, where administrative and medication-related tasks slow patient throughput. These constraints reduce the clinic's ability to fully utilize existing demand and limit overall revenue generation. This reduction in available clinical time directly limits the number of billable services that can be integrated into each patient encounter.

In addition to these structural constraints, internal scheduling data indicated a consistent pattern of cancelled appointments (Hometown Medical PLLC, 2026e). Although patient demand remained strong, missed appointments prevented scheduled visits from being converted into completed encounters, directly reducing potential revenue. Each cancelled visit represented lost clinical time that could not be recovered, reinforcing inefficiencies within the current workflow and further restricting available provider capacity. Collectively, these factors demonstrated that operational bottlenecks, rather than a lack of patient interest, were the primary obstacles to financial optimization.

(7) Synthesis of Findings

The combined financial and operational analysis demonstrates that revenue growth has not translated into improved profitability due to structural inefficiencies within the current operating model (Hometown Medical PLLC, 2026a, 2026c, 2026d, 2026e). The model remains highly dependent on patient volume, while reimbursement constraints, revenue leakage, and rising operating costs limit the clinic's ability to convert increased activity into sustainable financial performance. In addition, referral patterns indicate that a portion of revenue-generating services is currently performed outside the clinic, reducing total revenue per encounter.

Operational inefficiencies, including administrative workload and scheduling gaps, further constrain performance (Hometown Medical PLLC, 2026b, 2026c, 2026e). These findings reinforce the idea that improving revenue per patient encounter, rather than increasing patient volume, is the most viable pathway toward financial sustainability given the clinic’s current cost structure and reimbursement constraints.

(8) Key Implications

The findings indicate that Hometown Medical must transition from a volume-driven model to one that prioritizes revenue per patient encounter, service retention, and operational efficiency. This requires a targeted shift toward integrating higher-value services, reducing administrative inefficiencies, and capturing revenue currently lost through external referrals. This supports the evaluation of targeted service expansion strategies, such as the integration of in-house diagnostic capabilities, as a means of improving financial performance without relying on continued patient volume growth. These findings confirmed that the clinic’s primary constraint was not patient demand, but its ability to generate sufficient value within each patient encounter. This analysis was limited to a single-clinic dataset and may have been affected by variability in manually recorded operational data. As a result, the findings may not be fully generalizable to other independent primary care practices.

VII. IMPLEMENTATION

The findings from this analysis indicate that Hometown Medical’s current operating model is constrained not by patient demand, but by low revenue per patient encounter, high administrative costs, and significant referral leakage. Addressing these challenges requires targeted operational and service-level changes that improve revenue capture, increase efficiency,

and reduce reliance on external providers. The following recommendations focus on service expansion, workflow optimization, and revenue cycle improvements to support a more sustainable operating model.

(1) In-House Service Expansion (POCUS + Value-Based Care)

The integration of POCUS offers a strategic path to expand diagnostic capabilities while allowing the clinic to capture revenue that is currently leaking to external providers. According to the 2026 CMS Physician Fee Schedule, performing these ultrasounds in a non-facility setting can increase the total reimbursement per encounter when integrated with primary care services. While the initial investment for a high-performance device and provider training is estimated at \$12,000 to \$17,000, the business case is strong (Department of Family Medicine, 2026). By targeting a mid-range investment of \$14,000 and accounting for \$500 in annual variable costs (supplies and maintenance), the clinic can achieve a sustainable ROI. Based on internal referral patterns, capturing 40% (approximately 32 scans annually) of MSK and abdominal scans as add-on services to existing encounters could generate \$7,600 in annual revenue. This approach allows the equipment to achieve a full return on investment within 24 months without reducing standard billable office visits.

Beyond diagnostic imaging, the clinic can further stabilize its finances by prioritizing Medicare-specific service lines that maximize the value of the existing patient base. For instance, launching a Chronic Care Management (CCM) program allows the clinic to generate recurring revenue through non-face-to-face care coordination. Scaling this to 50 eligible patients could generate approximately \$38,000 in annual revenue, depending on patient enrollment and documentation compliance (Centers for Medicare & Medicaid Services, 2024). These recurring

revenue streams can be further supported through structured preventive care visits that increase patient engagement and improve reimbursement stability.

To facilitate these programs, the clinic should utilize Annual Wellness Visits (AWV) as a strategic point of entry for patient enrollment. These preventive encounters are 100% covered by Medicare with no patient co-pay, which reduces financial barriers to scheduling and completion (Centers for Medicare & Medicaid Services, 2024). In addition, AWVs typically reimburse at higher or comparable rates than standard evaluation and management visits, providing a predictable revenue stream while simultaneously improving quality metrics. Together, these service expansions move the clinic away from a volume-dependent model toward a more resilient, value-driven strategy.

(2) Workflow Optimization

To facilitate service expansion without escalating provider workload or contributing to burnout, the clinic must prioritize the resolution of administrative inefficiencies. A primary opportunity for improvement is the delegation of prior authorization responsibilities, as providers often allocate a significant portion of their time to these non-clinical administrative tasks (American Medical Association, 2024). By cross-training existing on-site clinical staff to manage the authorization process, the provider can reallocate this professional time toward direct patient care and higher-value diagnostic services. This approach uses existing staff capacity to reduce administrative bottlenecks without requiring additional hiring. Based on standard 20-minute appointment intervals, this shift could increase clinical capacity by approximately 8 to 12 additional patient encounters per week. At current reimbursement benchmarks, this increased throughput could generate an estimated \$800 to \$1,200 in incremental weekly revenue, depending on the specific payer mix and visit type.

Collectively, these strategic adjustments enhance the efficiency of the existing staffing model at Hometown Medical PLLC. By streamlining these administrative workflows, the clinic can expand its service offerings and maximize financial sustainability while maintaining its commitment to high-quality patient care. Ultimately, reducing manual labor allows the staff to prioritize patient-facing interactions, ensuring that operational growth does not compromise the clinic's clinical autonomy or patient experience.

(3) Revenue Cycle Improvements

To close the gap between billed charges and realized revenue, the clinic could implement targeted improvements to billing and collection processes. Financial reports from the first quarter of 2026 indicate a Net Collection Rate of approximately 87 percent. While this is below the commonly cited benchmark range, it reflects the clinic's specific payer mix, which includes a service area with a 14 percent poverty rate and a high volume of patients who are either covered by Medicaid or uninsured. By implementing monthly audits, the clinic can shift toward a proactive model that identifies errors in registration, coding, and coverage verification at the source. The goal is to reach a benchmark range of 95 to 98 percent, acknowledging the inherent limitations of the local economic context while still outperforming current baseline figures (Medical Group Management Association, 2026).

Furthermore, formalizing the workflow for patient-responsible balances will help stabilize cash flow. In addition to a structured three-notice system, the clinic should evaluate a Credit Card on File (CCOF) policy for small balances to reduce the administrative labor associated with manual collections. This systematic approach ensures that outstanding balances are recovered more efficiently while maintaining a patient-centered and flexible payment approach.

(4) Expected Impact

The combined implementation of in-house diagnostic services, workflow optimization, and revenue cycle improvements is expected to increase the revenue generated per encounter while minimizing operational waste. By capturing previously referred services and reallocating provider time to direct patient care, the clinic could improve its current 87 percent net collection rate and move closer to the national best-practice range of 95 to 98 percent (Medical Group Management Association, 2026). This transition from a volume-dependent model to a stable, value-driven operating structure provides a practical pathway toward long-term financial sustainability in an increasingly complex regulatory environment. These strategic adjustments position Hometown Medical PLLC to remain competitive and financially resilient without compromising the quality of patient care or clinical autonomy.

VIII. PROJECT EVALUATION

a. Project Outcome

This project identified that Hometown Medical PLLC's financial sustainability is hindered not by a lack of demand, but by structural limitations such as low revenue per encounter, significant referral leakage, and high administrative overhead. The project's analysis confirmed that increasing patient volume is no longer a viable strategy for improving profitability without a simultaneous shift toward service diversification and workflow optimization. Ultimately, the proposed implementation framework establishes a data-driven pathway for the clinic to transition into a resilient, value-driven model that ensures long-term viability without compromising clinical quality.

b. Contribution to Company

This project provided Hometown Medical PLLC with a strategic framework to improve financial performance within a constrained reimbursement environment. By quantifying the impact of referral leakage, the analysis established a clear business case for investing in in-house diagnostic capabilities and value-based service lines. These insights support informed decision-making and establish a scalable implementation plan that has the potential to allow the clinic to maintain its clinical independence while serving its patient population effectively. The framework also offers a practical roadmap for phased implementation, allowing leadership to align financial improvements with existing staffing capacity and patient care priorities.

c. Contribution to Student

This capstone project provided an opportunity to apply advanced analytics techniques to a real-world healthcare problem while strengthening skills in data preparation, integration, and analysis using administrative datasets. It reinforced how prior coursework translates into data-driven decision-making, particularly in evaluating financial performance, identifying revenue leakage, and assessing operational efficiency. The project also strengthened the ability to translate complex analytical findings into actionable recommendations for nontechnical stakeholders. Ultimately, the experience prepared the students for analytics-focused roles where data-informed decision-making is essential for improving organizational performance in healthcare settings.

d. Performance of Team

Our collaboration strengthened our ability to operate as a cohesive, high-performing project team. We maintained professionalism and adaptability while addressing the complexities of integrating internal clinic data with external benchmark and reimbursement datasets. Microsoft

Teams served as our primary collaboration platform for coordinating tasks, maintaining a shared repository for project documentation, and tracking updates.

Responsibilities were assigned based on individual strengths, including project management, data integration and processing, and statistical modeling. This division of labor improved efficiency and supported steady progress toward key milestones. Regular check-ins and a shared commitment to reproducibility kept the workflow transparent and aligned. When challenges emerged, we focused on identifying solutions while maintaining clear, respectful communication. The final product reflects both strong individual contributions and the value of a collective effort to produce actionable healthcare analytics.

IX. CONCLUSION

This project demonstrates that Hometown Medical PLLC's pathway to financial sustainability depends on transitioning from a volume-dependent model to a value-driven strategy. The analysis confirms that increasing patient volume alone does not translate into improved profitability due to reimbursement constraints, administrative burden, and rising operating costs. Instead, financial performance appears limited by the clinic's ability to generate sufficient revenue per patient encounter.

By quantifying the impact of revenue leakage, margin compression, and operational inefficiencies, the findings established a clear case for targeted service expansion and workflow optimization. The integration of in-house diagnostic capabilities, combined with improved administrative efficiency and stronger revenue cycle management, provided a practical and data-driven framework for improving performance. These combined strategies offer a scalable approach to optimizing existing resources while mitigating the financial risks associated with the current reimbursement landscape.

The proposed strategic shift supports the clinic in remaining both a viable business and a critical resource for the community it serves. This comprehensive transition aims to promote long-term financial resilience while allowing the practice to maintain its clinical independence and commitment to patient care. Ultimately, these improvements provide a framework that could help the clinic adapt to a shifting healthcare landscape without compromising its core mission. More importantly, the findings demonstrate that small, independent practices can strengthen financial performance through targeted operational changes rather than relying solely on increased patient volume.

X. APPENDIX

a. Figures 1 through 5

Figure 1: *Revenue vs Net Income*

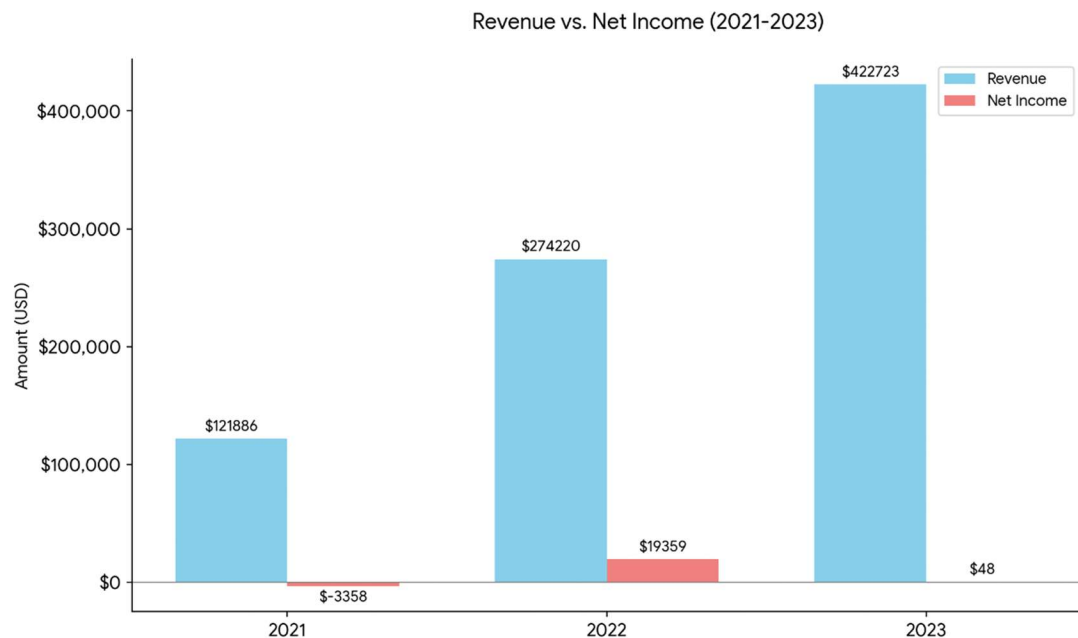


Figure 2: *Referral Leakage Chart*

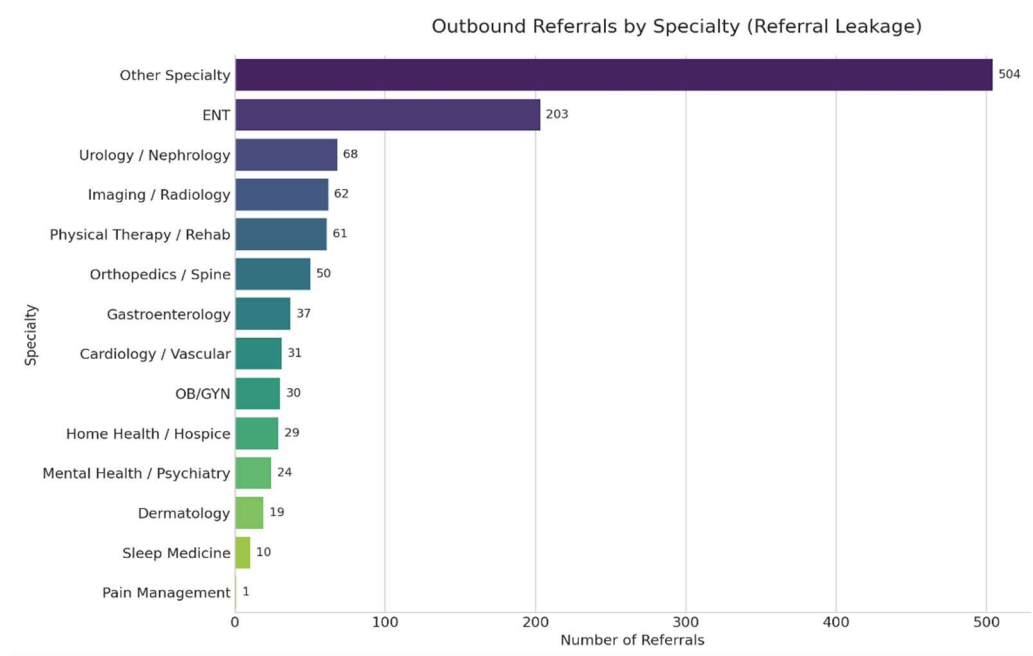


Figure 3: *Visits Per Day Distribution*

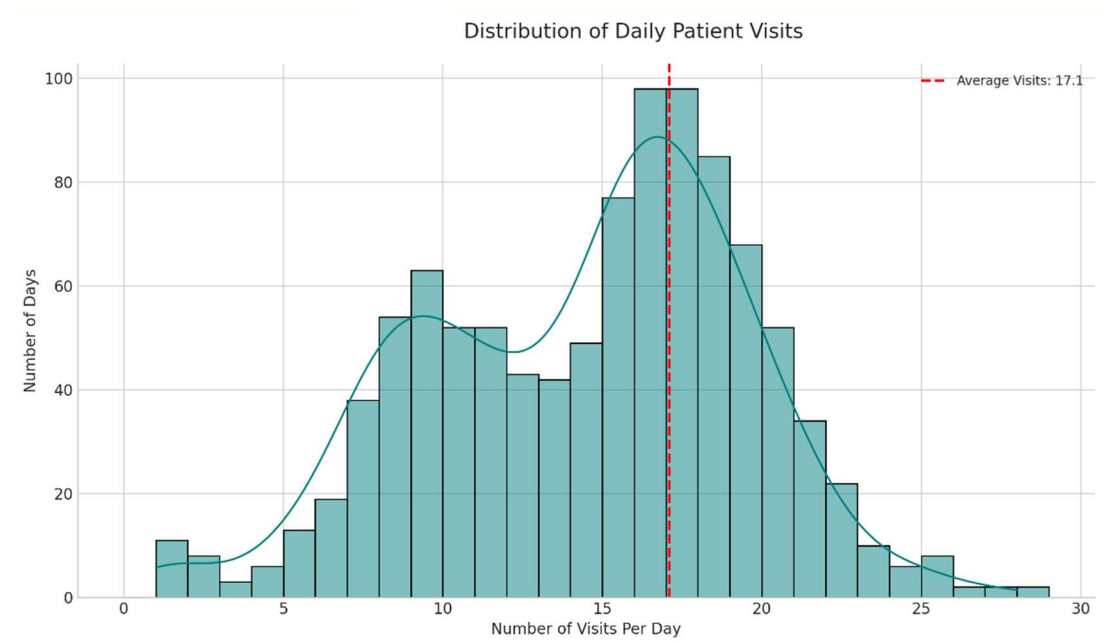


Figure 4: *Procedure / Revenue Opportunity Chart*

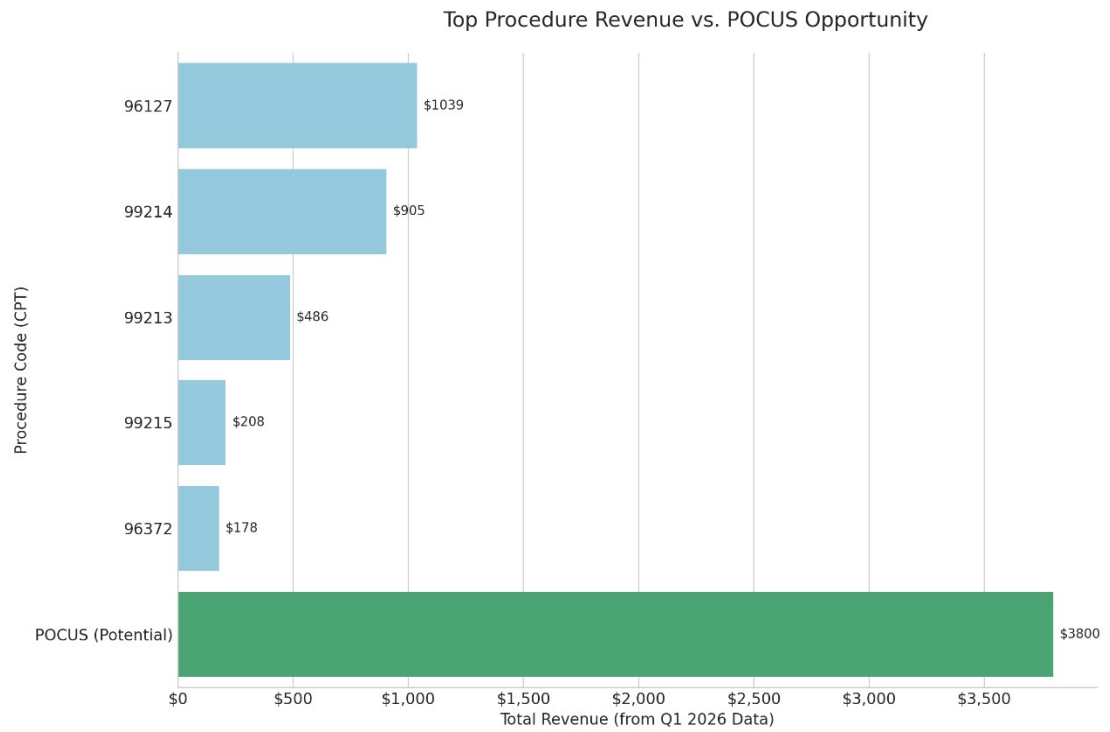
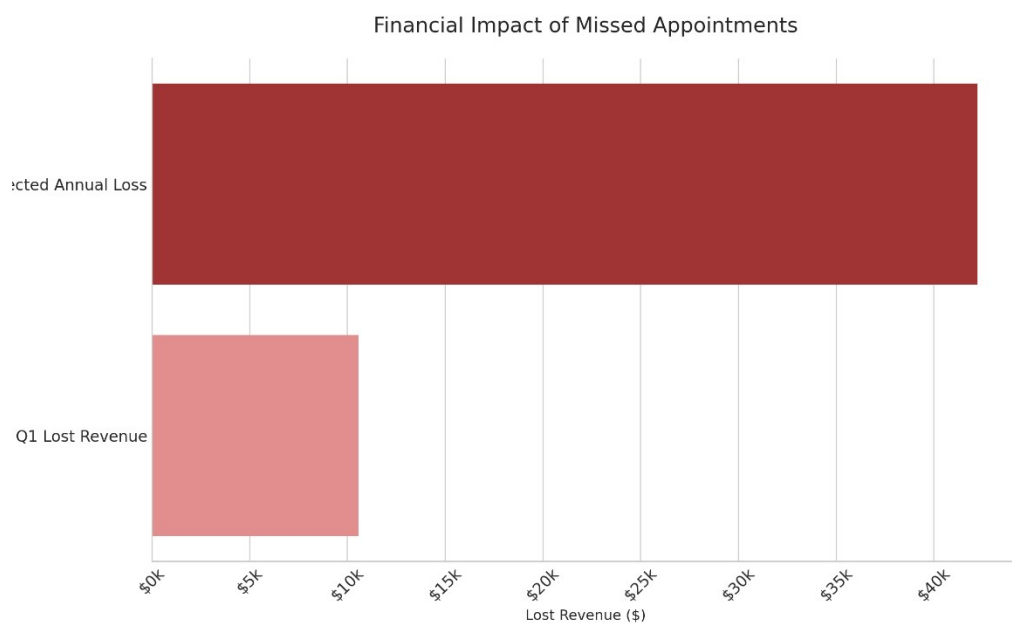


Figure 5: *Financial Impact of Missed Appointments*



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